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Degree : B.Tech.Semester:III-Course Work
END SEMESTER EXAMINATION (SUMMER),July,2026
Course Code :ECMTC301
Course Title:Mathematics for Communication and Signal Processing

Time:03 hours

Marks:50

Note: Attempt all five questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO
Q1	Attempt any 2 parts of the following:		
1(a)	Let $G = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in \mathbb{Z} \right\}$ be a group under matrix addition. Let $H = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in G : 2a - 3d = 0 \right\}$ Prove that H is a subgroup of G. What happens if the condition $2a - 3d = 0$ is replaced by $2a - 3d = 2$?	5	CO1
1(b)	Let $(\mathbb{R}, +)$ and $(\mathbb{R}^+, \cdot)$ denote the additive and multiplicative groups, respectively. Show that the map $\phi: \mathbb{R} \rightarrow \mathbb{R}^+$ given by $\phi(x) = e^x$ is a group homomorphism. Also, determine the kernel of ϕ .	5	CO1
1(c)	Define the term field . Determine whether the ring $\mathbb{Z}_{11}$ is a field. Justify your answer. Also comment on whether the ring $\mathbb{Z}_{12}$ is a field or not.	5	CO1
Q2	Attempt any 2 parts of the following:		
2(a)	Define the term subspace of a vector space. Let $W = \{(x, y, z) \in \mathbb{R}^3 : x - 2y + 3z = 0\}$. Determine whether W is a subspace of $\mathbb{R}^3$. Justify your answer.	5	CO2
2(b)	Define linearly independent and linearly dependent vectors. Determine whether the vectors $(1, 0, 2), (2, 1, 3), (3, 1, 5)$ form a basis for $\mathbb{R}^3$ over $\mathbb{R}$.	5	CO2
2(c)	Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, y - z)$. Show that T is a linear transformation. Determine $\dim(\ker T)$ and hence check the invertibility of T	5	CO2

Q3	Attempt any 2 parts of the following:		
3(a)	If the directional derivative of $\phi = axy + byz + czx$ at (1,1,1) has maximum magnitude 4 in a direction parallel to x axis, find the values of a,b,c.	5	CO3
3(b)	For constant vector $\vec{a}$, show that (i) $\nabla(\vec{a} \cdot \vec{r}) = \vec{a}$ (ii) $\nabla \times (\vec{a} \times \vec{r}) = 2\vec{a}$	5	CO3
3(c)	Verify Green's theorem for the field $\vec{F} = x^2\vec{i} + xy\vec{j}$ over the region R enclosed by $y = x^2$ and the line $y = x$.	5	CO3
Q4	Attempt any 2 parts of the following:		
4(a)	Obtain Laplace transform of $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 4y + 2\int_0^t y(t)dt$	5	CO4
4(b)	Use the Convolution theorem to find inverse Laplace Transform of $\frac{1}{s^2(s+1)^2}$	5	CO4
4(c)	If $f(t) = \begin{cases} t+1 & , 0 \leq t \leq 2 \\ 3 & , t > 2 \end{cases}$ Find $L[f(t)]$	5	CO4
Q5	Attempt any 2 parts of the following:		
5(a)	A homogeneous rod of conducting material of length 100 cm has its ends kept at zero temperature and the temperature initially is $U(x,0) = \begin{cases} x & , 0 \leq x \leq 50 \\ 100 - x & , 50 \leq x \leq 100 \end{cases}$ Find the temperature $u(x,t)$ at any time t.	5	CO5
5(b)	A string is stretched and fastened to two points L apart. Motion is started by displacing the string in the form $u = a \sin \frac{\pi x}{L}$ from which it is released at time $t = 0$. Find the displacement $u(x,t)$ from one end.	5	CO5
5(c)	Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ which satisfies the conditions $u(0,y) = u(l,y) = u(x,0) = 0$ and $u(x,a) = \sin\left(\frac{n\pi x}{l}\right)$.	5	CO5